Dictionary Graph Topology as a Probe of LLM Conceptual Structure: Kernel Contraction and Cycle Irreducibility Scale with Model Size

Gendo Kumoi Nagaoka University of Technology, Japan kumoi@vos.nagaokaut.ac.jp

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Abstract

We construct directed definitional graphs from definitions generated by Large Language Models (LLMs) and apply the graph-theoretic framework of Vincent-Lamarre et al. (2016) to quantify lexical self-sufficiency and conceptual circularity. Across 3,000 common English words and seven models spanning two families (Qwen3.5: 0.8B – 27B; Gemma-4: 4B – 31B), we find that the *kernel ratio* (fraction of vocabulary in the minimal self-referential core) decreases monotonically with model size within the Qwen3.5 family (Pearson $r = -0.862$), while the *MinSet/Kernel ratio* (fraction of the kernel that is irreducibly cyclic) increases monotonically ($r = +0.982$). We identify 92 words present in the kernel of every Qwen3.5 model—a universal lexical anchor—of which 10 overlap with Wierzbicka's Natural Semantic Metalanguage (NSM) primitives. A cross-family comparison reveals that instruction-following ability (here, avoidance of self-referential definitions) is a family-level property, not a scale property: Qwen3.5 models define words using the word itself 76 – 92% of the time regardless of scale, whereas Gemma-4 does so less than 0.1% of the time. We treat this self-referential rate as an *instruction-regime covariate* and show that, after partialling it out, kernel-ratio remains a robust probe of scale (r_partial $= -0.723$). Graph metrics are stable across 4-bit, 6-bit, and 8-bit quantization of the same model (kernel-ratio variance < 1 pp). These results suggest that definitional graph topology provides a scale-sensitive, quantization-robust window into how LLMs organise conceptual knowledge.

Keywords: large language models, definitional graphs, kernel, conceptual circularity, scaling laws, instruction following

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1 Introduction

How do large language models organise conceptual knowledge, and does that organisation change with scale? The most direct evidence of conceptual structure is a model's ability to *define* words—to place a term in relation to other terms. The resulting network of definitions carries information that is invisible in standard benchmark evaluations: which words are fundamentally explanatory, which form irreducible cycles of mutual dependence, and how tightly connected the conceptual fabric is.

Formal analysis of dictionary graphs dates to Blondin Massé et al. (2008) and was systematised by Vincent-Lamarre et al. (2016), who measured the *Kernel* (the minimal set of words whose definitions stay within the set), the *Core* (the set of source strongly-connected components in the Kernel subgraph), and the *MinSet* (the minimum feedback vertex set of the Kernel—i.e., the smallest set of words whose removal breaks all cycles). For four major English dictionaries, the Kernel contains roughly 10% of the vocabulary, the Core accounts for 65 – 90% of the Kernel, and the MinSet is about 1% of the Kernel. These ratios characterise a dictionary' s *semantic closure.*

We ask: do LLMs exhibit semantic closure, and does its character change with model size? This question has practical and theoretical stakes. Practically, a model whose definitions form a large, tightly cyclic Kernel may be "going in circles" —defining terms by synonyms without ever grounding them. Theoretically, if graph topology tracks parameter count, it constitutes a new dimension of the scaling-law literature, complementing loss-based and benchmark-based metrics.

Our contributions are:

1. We construct and analyse definitional graphs for seven LLMs from two families across 3,000 words.
2. We demonstrate a robust monotone relationship between kernel ratio and log-parameter count within the Qwen3.5 family.
3. We discover 92 scale-invariant kernel words—a universal lexical anchor—with a 15.4% overlap with NSM semantic primitives.
4. We characterise instruction-following ability (avoidance of self-referential definitions) as a family-level property and show it acts as a suppressor variable in cross-family regression.
5. We demonstrate quantization-robustness of graph metrics down to 4-bit precision.

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2 Background and Related Work

2.1 Dictionary Graph Theory

Following Vincent-Lamarre et al. (2016), a *dictionary graph* is a directed graph $G = (V, E)$ where V is the set of entry words and $(u, v) \in E$ if and only if word u appears in the definition of word v. The direction encodes "u is used to define v." Key constructs:

- **Kernel**: The subgraph remaining after iteratively removing nodes with out-degree 0. Each removal step eliminates words that do not appear in any surviving definition. The result is the minimal set closed under "being used to define" — the vocabulary sufficient to express all other definitions.
- **Core**: The union of source SCCs (strongly connected components with no incoming arcs in the condensation DAG) within the Kernel.

- **MinSet / Minimum Feedback Vertex Set (MFVS)**: The smallest set of words whose removal renders the Kernel acyclic. Computed via integer linear programming (PuLP + HiGHS solver).

For human dictionaries, these metrics reveal that lexicons are not arbitrarily self-referential: a small, tightly connected core bootstraps the rest of the vocabulary (Harnad et al., 2013).

2.2 LLMs and Graph-Structured Knowledge

Recent work has explored whether LLMs possess graph-reasoning capabilities. "Are LLMs Truly Graph-Savvy?" (ACL SRW 2025) shows LLMs struggle with structural graph tasks when token-level shortcuts are removed. GraCoRe (COLING 2025) introduces a benchmark for grounded compositional reasoning over graph structures. These results concern *reasoning about* explicit graphs, whereas our approach uses implicit graph structure in *generated* definitions as a probe.

"From Tokens to Thoughts" (2025) examines whether LLM-induced semantic spaces compress concepts similarly to human lexical spaces, finding systematic differences tied to training data and architecture.

2.3 Scaling Laws and Knowledge Capacity

Scaling laws (Hoffmann et al., 2022; Kaplan et al., 2020) characterise how benchmark performance scales with parameters and training compute. Recent work (arXiv:2404.05405) estimates factual knowledge capacity at approximately 2 bits per parameter, with phase-transition-like emergence of structured knowledge at larger scales. Our graph-metric approach offers a complementary, architecture-agnostic probe that does not depend on factual recall but on relational structure.

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3 Method

3.1 Word Sampling

We sample 3,000 words (1,500 nouns, 900 verbs, 600 adjectives) from WordNet, filtered by Brown Corpus frequency (minimum frequency 5) to ensure words commonly appear in cross-definitions. Words are sampled with random seed 42.

3.2 Definition Generation

Each model generates one definition per word using the prompt:

> *Define the {POS}* "*{word}*" *in one short sentence. Use only common English words. Do not use the word itself in the definition.*

Generation parameters follow Qwen3.5' s official non-thinking mode recommendation: temperature = 0.7, top-p = 0.8, top-k = 20, max tokens = 80. Inference is served locally via mlx-proxy (Apple M4 Max). Definitions where the target word appears verbatim are flagged as `self_referential` (status preserved for graph construction but recorded as the *sr_rate* covariate).

3.3 Graph Construction

Definitions are tokenised, lowercased, and lemmatised with WordNetLemmatizer. Stopwords and out-of-vocabulary tokens are removed. For each definition of word v, edges (u, v) are added for every lemma u in the definition. Self-loops are removed. Both `ok` and `self_referential` definitions contribute edges (self-loops from the latter are excluded). The resulting graph has 2,750 nodes (250 words failed generation across models).

3.4 Metrics

- **kernel_ratio**: |Kernel| / |V|
- **circulation_rate**: fraction of nodes in SCCs of size ≥ 2
- **mean_out_degree**: mean number of definitional usages per word
- **mset/k** (MinSet/Kernel ratio): |MinSet| / |Kernel|
- **sr_rate**: fraction of definitions flagged as self-referential

A WordNet baseline is constructed from `synset.definition()` outputs passed through the identical pipeline.

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4 Results

4.1 Models and Data

Table 1: All models — graph metrics and instruction-following rate

Model	Family	Params	Follow%	kern%	mset/k%	circ%	out_deg
Qwen3.5-0.8B	Qwen3.5	0.8B	23.8	17.9	14.4	16.7	3.73
Qwen3.5-2B	Qwen3.5	2B	7.8	11.1	16.7	6.1	2.66
Qwen3.5-4B	Qwen3.5	4B	13.8	10.8	17.1	7.5	2.49
Qwen3.5-9B	Qwen3.5	9B	24.1	10.1	18.1	7.6	2.57
Qwen3.5-27B	Qwen3.5	27B	15.1	8.6	20.3	6.5	2.52

Model	Family	Params	Follow%	kern%	mset/k%	circ%	out_deg
Gemma-4-4B	Gemma-4	4B	**99.9**	7.3	23.3	4.3	—
Gemma-4-31B	Gemma-4	31B	**99.9**	8.1	16.2	5.1	—
WordNet (baseline)	—	—	—	15.5	—	7.4	1.67

Figure note: Kernel ratio decreases with parameter count within Qwen3.5 ($r = -0.862$). Gemma-4 achieves lower kernel ratios than same-size Qwen models despite 99.9% instruction-following.

4.2 Scaling Relationship within Qwen3.5

Within the Qwen3.5 family, graph metrics exhibit strong monotone relationships with $\log_2(\text{params})$:

- **kernel_ratio** decreases: $r = -0.862$ ($p < 0.05$)
- **mset/k** increases: $r = +0.982$ ($p < 0.005$)
- **circulation_rate** decreases: $r = -0.675$

The kernel contracts as model size grows: larger models require a smaller fraction of the vocabulary to bootstrap definitions. Simultaneously, the irreducible cyclic fraction of that kernel *increases*—larger models have denser, more tightly interlocked core concepts.

OLS regression (kernel_ratio ~ $\log_2(\text{params})$ + sr_rate):

$$\text{kernel\_ratio} = -0.0165 \cdot \log_2(\text{B}) - 0.0003 \cdot \text{sr\_rate} + 0.2003$$

$R^2 = 0.751$. Adding sr_rate to the regression has minimal impact ($\Delta r = +0.039$ for partial correlation), confirming that the kernel-ratio – scale relationship is genuine, not a byproduct of differential instruction-following.

4.3 Universal Kernel: 92 Scale-Invariant Words

Ninety-two words appear in the kernel of every Qwen3.5 model, regardless of parameter count. These include basic action verbs (*go*, *hold*, *move*, *use*), relational adjectives (*different*, *flat*, *high*, *small*), and cognitive primitives (*feel*, *idea*, *image*, *true*). Ten of the 92 overlap with Wierzbicka' s (2021) Natural Semantic Metalanguage primitives: *body*, *feel*, *like*, *people*, *place*, *small*, *time*, *touch*, *true*, *two*.

This universal kernel is consistent across all model sizes despite a 34× difference in parameter count, suggesting that the definitional role of these concepts is determined by English lexical structure, not by LLM scale.

4.4 Cross-Family: Instruction-Regime as Structural Determinant

Gemma-4 follows the “do not use the word itself” instruction nearly perfectly (sr_rate < 0.1%), whereas Qwen3.5 ignores it in 76 – 92% of definitions. This is a **family-level property**, not a function of scale: Qwen3.5 shows no monotone improvement in instruction-following across the 0.8B – 27B range.

The structural consequence is profound. Gemma-4-4B produces sparser graphs (5,025 edges vs. 6,844 for same-size Qwen3.5-4B) and a lower kernel ratio (7.3% vs. 10.8%). However, Gemma-4-4B’ s mset/k ratio is *higher* (23.3% vs. 17.1%), indicating that its smaller kernel is more cyclically irreducible.

Partial correlation analysis (controlling for sr_rate):

Metric	r_raw	r_partial|sr	Δ r	Interpretation
kernel_ratio	− 0.757	− 0.723	+0.034	sr has little effect
mset/k	+0.318	+0.388	+0.070	sr was mildly suppressing
circ	− 0.436	− 0.526	− 0.090	sr was mildly inflating

We recommend reporting sr_rate as an *instruction-regime covariate* in cross-family comparisons.

4.5 Quantization Robustness

Applying 4-bit, 6-bit, and 8-bit quantization to Qwen3.5-27B:

Precision	kern%	mset/k%	circ%
bf16 (reference)	8.6	20.3	6.5
8-bit	9.4	15.9	5.4
6-bit	8.2	19.1	4.5
4-bit	8.7	20.2	6.5

Kernel ratio varies by at most 0.8 pp across quantization levels. The 4-bit model is closest to bf16 on both kernel and mset/k metrics. The 8-bit model shows the largest deviation (Δ mset/k = − 4.4 pp), a potential artifact of the MLX 8-bit quantization scheme. Graph structure is largely quantization-robust at the kernel level.

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5 Discussion

5.1 Kernel Contraction as Lexical Efficiency

A decreasing kernel ratio with scale can be interpreted as *increasing lexical efficiency*: larger models rely on a smaller, more central set of anchor words to bootstrap their definitional vocabulary. This is consistent with the hypothesis that scale enables more precise semantic discrimination—larger models can define "table" without resorting to "flat surface" (itself requiring further definition), using instead a more specific conceptual vocabulary.

5.2 Increasing Cyclic Irreducibility

The increasing mset/k ratio with scale suggests that, while the kernel contracts, the remaining cyclic structure becomes *harder to break*. This may reflect that larger models encode denser, more mutually reinforcing conceptual clusters. Small models exhibit loose cycles that can be linearised by removing few words; large models' kernels are more topologically entangled.

5.3 Comparison with Human Dictionaries

WordNet (our baseline) yields kernel_ratio = 15.5%—higher than any LLM. This aligns with Vincent-Lamarre et al.'s finding of ~10% for curated human dictionaries (with WordNet's higher figure attributable to its synset-based, rather than prose, definitions). Larger LLMs converge toward the human-dictionary range (Qwen3.5-27B: 8.6%), though no model yet achieves the MinSet economy observed in LDOCE or Cambridge.

5.4 NSM Alignment

The 15.4% NSM overlap in the universal kernel (10/65 primitives) is non-trivial but below the ~75% overlap reported for human dictionary cores in Vincent-Lamarre et al. This gap may reflect the different nature of LLM vs. human definitional practice: LLMs generate functional descriptions, not minimal semantic decompositions.

5.5 Limitations

- **Five data points** (Qwen3.5) limit statistical power. Cross-family comparison is confounded by instruction-regime.
- **Prompt sensitivity**: The self-referential rate is highly prompt-dependent; different prompts may elicit different regime behaviour.
- **Single seed**: Stochastic generation at T=0.7 introduces variance not captured here.
- **WordNet only** as human baseline; curated dictionaries (LDOCE, Cambridge) would enable closer comparison to Vincent-Lamarre et al.

6 Conclusion

Definitional graph topology provides a scale-sensitive, quantization-robust lens on LLM conceptual structure. Within the Qwen3.5 family, kernel ratio decreases and MinSet/Kernel ratio increases monotonically with model size, suggesting that larger models organise their definitional vocabulary more efficiently around a denser conceptual core. Ninety-two words form a scale-invariant lexical anchor, and a 15.4% overlap with NSM primitives hints at universality. Cross-family comparisons must account for the instruction-following regime: Gemma-4' s near-zero self-referential rate constitutes a fundamentally different definitional mode that cannot be compared directly without controlling for sr_rate. These results establish dictionary graph analysis as a viable complement to loss-based scaling laws for characterising the internal knowledge organisation of language models.

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